

# Dataset Usage Summary

## Quick Answer

**Dataset Used:** Synthetic Demo Dataset (generated programmatically)

**Location:** data/processed/synthetic\_demo/metadata.jsonl

**Size:** 1,000 samples (configurable)

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## Files That Use the Dataset

### 1. ml/data\_ingest.py (GENERATES THE DATASET)

- **Function:** `generate_synthetic_samples(count, output_dir)`
- **Lines:** 70-140
- **Purpose:** Creates synthetic training data with realistic gaze angles and attention labels
- **Output:**
  - data/processed/synthetic\_demo/metadata.jsonl (metadata)
  - data/processed/synthetic\_demo/frames/\*.jpg (images)

### 2. ml/train.py (TRAINS THE MODEL)

- **Function:** `main()`
- **Lines:** 60-120
- **Purpose:** Loads dataset and trains the attention detection model
- **Key Operations:**

```
samples = load_samples(cfg.data_path)  # Reads metadata.jsonl
train_samples, val_samples = train_val_split(samples)
train_data = preprocess_samples(train_samples, cfg.image_size)
model.fit(train_data["images"], train_data["targets"], ...)
```

### 3. ml/dataset.py (LOADS AND PROCESSES DATA)

- **Functions:**
  - `load_samples(metadata_path)` - Reads JSONL file
  - `train_val_split(samples)` - Splits into train/validation
  - `preprocess_samples(samples)` - Converts to model format
- **Purpose:** Provides data loading utilities for training

### 4. ml/eval.py (EVALUATES MODEL)

- **Purpose:** Tests model performance on validation dataset
  - **Uses:** Same dataset loading functions from `ml/dataset.py`
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## How to Generate the Dataset

```
# Generate 1000 synthetic samples
python ml/data_ingest.py ingest synthetic_demo --limit 1000

# Output:
# - data/processed/synthetic_demo/metadata.jsonl
# - data/processed/synthetic_demo/frames/synthetic_00000.jpg
# - data/processed/synthetic_demo/frames/synthetic_00001.jpg
# - ... (1000 images total)
```

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## How to Train with the Dataset

```
# Train model using the generated dataset
python ml/train.py \
  --data data/processed/synthetic_demo/metadata.jsonl \
  --epochs 10 \
  --batch-size 32

# Output:
# - artifacts/focusmate_model/saved_model/ (trained model)
# - artifacts/focusmate_model/checkpoints/ (training checkpoints)
# - artifacts/focusmate_model/logs/ (TensorBoard logs)
```

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## Dataset Format

Each line in metadata.jsonl:

```
{
  "frame_path": "frames/synthetic_00000.jpg",
  "gaze_yaw": -15.76,
  "gaze_pitch": -4.40,
  "head_yaw": -6.96,
  "head_pitch": -1.15,
  "eyes_open_prob": 0.788,
  "label": "looking_left",
  "confidence": 0.789,
  "meta": {"source": "synthetic", "index": 0}
}
```

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## 11 Attention Labels

1. focused - Looking at screen
2. looking\_left - Gaze directed left
3. looking\_right - Gaze directed right
4. looking\_up - Gaze directed upward
5. looking\_down - Gaze directed downward
6. engaged - Generally attentive

7. `partial_engaged` - Partially attentive
  8. `sleepy` - Eyes closing
  9. `distracted_by_multi_face` - Multiple faces
  10. `no_face` - No face detected
  11. `unknown` - Uncertain state
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## Complete Training Pipeline

```
# Step 1: Generate dataset
python ml/data_ingest.py ingest synthetic_demo --limit 1000

# Step 2: Train model
python ml/train.py --data data/processed/synthetic_demo/metadata.jsonl

# Step 3: Export for browser
python ml/export_model.py \
  --model artifacts/focusmate_model/saved_model \
  --out frontend/public/models/focusmate-attention \
  --format tfjs
```

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### Method 1: Using Pandoc

```
pandoc DATASET_DOCUMENTATION.md -o DATASET_DOCUMENTATION.pdf
```

### Method 2: Using Markdown to PDF Online

1. Open <https://www.markdowntopdf.com/>
2. Upload `DATASET_DOCUMENTATION.md`
3. Download the PDF

### Method 3: Using VS Code Extension

1. Install "Markdown PDF" extension
2. Open `DATASET_DOCUMENTATION.md`
3. Right-click → "Markdown PDF: Export (pdf)"